

set up yunikorn queue

set up some child queues under a parent queue with the same priority, e.g. a parent queue and 2 child queues like this:

The screenshot shows the YuniKorn web interface at `localhost:9889/#/queues`. The left sidebar contains navigation links: Dashboard, Queues (selected), Applications, Nodes, and Status. The main area is titled 'Queues' and has a 'Partition: default' dropdown. A tree view shows the hierarchy: 'root' (expanded) contains 'root.default' and 'root.ns-a' (expanded). 'root.ns-a' contains 'root.ns-a.tenant-a' and 'root.ns-a.tenant-b'. The 'Queue Info' panel for 'root.ns-a' is open on the right, showing the following details:

Queue Info	
Name:	root.ns-a
Status:	Active
Allocated:	n/a
Pending:	n/a
Max:	Memory: 5.86 GiB CPU: 1.2
Guaranteed:	n/a
Absolute Used Capacity:	n/a
Max Running Apps:	0
Running Apps:	0
preemption.delay:	10s
preemption.policy:	fence

The screenshot shows the YuniKorn web interface at `localhost:9889/#/queues`. The left sidebar contains navigation links: Dashboard, Queues (selected), Applications, Nodes, and Status. The main area is titled 'Queues' and has a 'Partition: default' dropdown. A tree view shows the hierarchy: 'root' (expanded) contains 'root.default' and 'root.ns-a' (expanded). 'root.ns-a' contains 'root.ns-a.tenant-a' and 'root.ns-a.tenant-b'. The 'Queue Info' panel for 'root.ns-a.tenant-a' is open on the right, showing the following details:

Queue Info	
Name:	root.ns-a.tenant-a
Status:	Active
Allocated:	n/a
Pending:	n/a
Max:	Memory: 5.86 GiB CPU: 1.2
Guaranteed:	Memory: 2.93 GiB CPU: 600m
Absolute Used Capacity:	n/a
Max Running Apps:	0
Running Apps:	0
preemption.delay:	10s
priority.offset:	0
Applications:	Link

YuniKorn

Dashboard

Queues

Applications

Nodes

Status

Queues

Partition: default

root

root.default

root.ns-a

root.ns-a.tenant-a

root.ns-a.tenant-b

Queue Info

Name: root.ns-a.tenant-b

Status: Active

Allocated: n/a

Pending: n/a

Max: Memory: 5.86 GiB
CPU: 1.2

Guaranteed: Memory: 2.93 GiB
CPU: 600m

Absolute Used Capacity: n/a

Max Running Apps: 0

Running Apps: 0

preemption.delay: 10s

priority.offset: 0

Applications: [Link](#)

a simple yunikorn-config example

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: yunikorn-configs
  namespace: yunikorn
data:
  queues.yaml: |
    partitions:
      - name: default
        placementrules:
          - name: provided
          - name: fixed
            value: root.default
        queues:
          - name: root
            queues:
              - name: default
                submitacl: '*'
              - name: ns-a
                resources:
                  max:
                    {memory: 6000Mi, vcore: 1200m}
            properties:
              preemption.policy: fence
              preemption.delay: 10s
            queues:
              - name: tenant-a
                submitacl: '*'
                resources:
                  max:
                    {memory: 6000Mi, vcore: 1200m}
                  guaranteed:
                    {memory: 3000Mi, vcore: 600m}
            properties:
              priority.offset: "0"
              - name: tenant-b
                submitacl: '*'
                resources:
                  max:
                    {memory: 6000Mi, vcore: 1200m}
                  guaranteed:
```

```
{memory: 3000Mi, vcore: 600m}  
properties:  
  priority.offset: "0"
```

test long-running large and short-running small spark jobs

submit long-running large spark job and put it on a child queue, e.g.

```
argo submit spark-pi-workflow.yaml \  
  -p image=apache/spark:3.5.7 \  
  -p partitions=100000 \  
  -p namespace=yunikorn-spark-submit \  
  -p executors=6 \  
  -p queue=root.ns-a.tenant-a \  
  -p scheduler=yunikorn
```

submit short-running small spark job and put it on a child queue, e.g.

```
argo submit spark-pi-workflow.yaml \  
  -p image=apache/spark:3.5.7 \  
  -p partitions=1000 \  
  -p namespace=yunikorn-spark-submit \  
  -p executors=2 \  
  -p queue=root.ns-a.tenant-b \  
  -p scheduler=yunikorn
```

spark-wf example

```
apiVersion: argoproj.io/v1alpha1
kind: Workflow
metadata:
  generateName: spark-pi-wf-
spec:
  entrypoint: spark-pi
  arguments:
    parameters:
      - name: queue
        value: root.default # default value
      - name: image
        value: apache/spark:3.5.7
      - name: namespace
        value: yunikorn-spark-submit
      - name: partitions
        value: 10000
      - name: executors
        value: 1
      - name: scheduler
        value: default-scheduler
  templates:
    - name: spark-pi
      serviceAccountName: spark
      container:
        image: "{{workflow.parameters.image}}"
        command: ["/opt/spark/bin/spark-submit"]
        args:
          - --master
          - k8s://https://kubernetes.default.svc:443
          - --deploy-mode
          - cluster
          - --name
          - spark-pi
          - --conf
          - spark.kubernetes.container.image={{workflow.parameters.image}}
          - --conf
          - spark.kubernetes.namespace={{workflow.parameters.namespace}}
          - --conf
          - spark.kubernetes.authenticate.driver.serviceAccountName=spark
          - --conf
          - spark.executor.instances={{workflow.parameters.executors}}
```

```
- --conf
- spark.kubernetes.driver.request.cores=0.2
- --conf
- spark.kubernetes.driver.limit.cores=0.5
- --conf
- spark.driver.memory=1024m
- --conf
- spark.driver.memoryOverhead=200m
- --conf
- spark.kubernetes.executor.request.cores=0.2
- --conf
- spark.kubernetes.executor.limit.cores=0.5
- --conf
- spark.executor.memory=512m
- --conf
- spark.executor.memoryOverhead=100m
- --conf
- spark.kubernetes.scheduler.name={{workflow.parameters.scheduler}}
- --conf
- spark.kubernetes.driver.label.queue={{workflow.parameters.queue}}
- --conf
- spark.kubernetes.driver.label.yunikorn.apache.org/allow-preemption=false
- --conf
- spark.kubernetes.executor.label.queue={{workflow.parameters.queue}}
- local:///opt/spark/examples/src/main/python/pi.py
- "{{workflow.parameters.partitions}}"
```

note: set [spark.kubernetes.driver.label.yunikorn.apache.org/allow-preemption=false](#), to prevent the driver from being preempted.